Chapter 4

Strategic Form Games

Strategic form representation is the most extensively discussed representation for games in this book. In this chapter, we provide a number of illustrative examples to help gain an intuitive understanding of strategic form games. The examples involve finite games as well as infinite games. The games discussed include matching pennies, rock-paper-scissors, Bach or Stravinsky, student's coordination, prisoner's dilemma, company's dilemma, duopoly pricing, tragedy of the commons, bandwidth sharing, sealed bid auction, Pigou network routing game, and Braess paradox. The examples are fairly representative of the wide canvas of applications where game theoretic modeling is relevant.

4.1 Preliminaries

We have seen in Chapter 2 (Definition 2.1) that a strategic form game Γ is a tuple $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ where $N = \{1, 2, \ldots, n\}$ is a finite set of players; $S_1, S_2, \ldots, S_n$ are the strategy sets of the players; and $u_i : S_1 \times S_2 \times \cdots \times S_n \to \mathbb{R}$ for $i = 1, 2, \ldots, n$ are utility functions. We have seen in Chapter 3 that games in extensive form can be transformed into strategic form games by mapping contingent action plans into strategies. The phrases *strategic form games*, *strategic games*, and *normal form games* are synonymous. When there is no confusion, we use the notation $\Gamma = \langle N, (S_i), (u_i) \rangle$ for a strategic form game.

We denote by S, the set of all strategy profiles or strategy vectors, which is the Cartesian product $S_1 \times \cdots \times S_n$. A typical strategy profile is represented by $(s_1, \ldots, s_n)$ where s_i is the strategy of player i ($i = 1, \ldots, n$). We denote by S_{-i} the Cartesian product $S_1 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots S_n$ of strategy sets of all players other than player i. We denote by s_{-i} a typical strategy profile in S_{-i}. When we are focusing on a particular player i, a convenient way of representing a strategy profile is (s_i, s_{-i}) where $s_i \in S_i$ and $s_{-i} \in S_{-i}$.

The idea behind the strategic form representation is that a player's decision problem is to essentially choose a strategy that will counter most effectively the strategies adopted by the other players. Such a strategy is called a *best response strategy* which is formally defined as follows.

Definition 4.1 (Best Response Strategy). *Given a strategic form game* $\Gamma = \langle N, (S_i), (u_i) \rangle$ *and a strategy profile* $s_{-i} \in S_{-i}$*, we say* $s_i \in S_i$ *is a best response strategy of player* i *with respect to* s_{-i} *if* $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})\ \forall s'_i \in S_i$.

Given a strategy profile $s_{-i} \in S_{-i}$ of all players except player i, there could exist multiple best response strategies for player i.

In a strategic form game, each player is faced with the problem of choosing his best response strategy and the players can be thought of as simultaneously choosing their strategies from the respective sets $S_1, \ldots, S_n$.

Many interpretations have been provided in the game theory literature for strategic form games. We present here two interpretations provided by Osborne and Rubinstein [1].

In the first interpretation, a strategic form game is a model of an event that occurs only once. Each player knows the details of the game and the fact that all players are rational. The players choose their strategies simultaneously and independently. Each player is unaware of the choices being made by the other players.

In the second interpretation, a player is assumed to form an expectation of the other players' behavior on the basis of information about the way the game or a similar game was played in the past. A strategic form game models a sequence of plays of the game under the condition that there is no strategic link between the plays of the game. That is, a player who plays the game many times should only worry about his own instantaneous payoff and ignore the effects of his current action on the future behavior of the other players. A class of games called *repeated games* is relevant if there is a strategic link between plays of a game.

The extensive form representation, discussed in Chapter 3, is a more detailed representation than the strategic form representation. A given strategic form game could correspond to multiple extensive form games. The strategic form game suppresses the dynamics of the game but is more convenient for certain kinds of analysis of games. It is enough to work with strategic form for finding answers to many useful analysis questions.

4.2 Matching Pennies with Simultaneous Moves

We have already studied this game in Chapter 3. Recall that in this game, two players 1 and 2 put down their respective rupee coins, heads or tails up. If both the coins match (both heads or both tails), then player 2 pays one rupee to player 1. Otherwise, player 1 pays one rupee to player 2. Let us say A denotes *heads* and B denotes *tails*. It is easy to see that:

$$
\begin{aligned}
N &= \{1, 2\} \\
S_1 &= S_2 = \{A, B\} \\
S &= S_1 \times S_2 = \{(A, A), (A, B), (B, A), (B, B)\}
\end{aligned}
$$

The payoff matrix is given by

	2	
1	A	B
A	$1, -1$	$-1, 1$
B	$-1, 1$	$1, -1$

This is perhaps the simplest example of a two player game. This belongs to a class of games called two player zero-sum games (so called because the sum of utilities in every outcome is equal to zero). In the above game, it is easy to see that the best response strategy of player 1 is A (B) when player 2 plays A (B). On the other hand, the best response strategy of player 2 is A (B) when player 1 plays B (A).

To understand why a simple game such as the above is important to study, let us consider the following situation. There are two companies, call them 1 and 2. Each company is capable of producing two products A and B, but at any given time, a company can only produce one product, due to high setup and switchover costs. Company 1 is known to produce superior quality products but company 2 scores over company 1 in terms of marketing and advertising. If both the companies produce the same product (A or B), it turns out that company 1 makes all the profits and company 2 loses out, because of the superior quality of products produced by company 1. This is reflected in our model with a payoff of $+1$ for company 1 and a payoff of -1 for company 2, corresponding to the strategy profiles (A, A) and (B, B).

On the other hand, if one company produces product A and the other produces product B, it may turn out that because of aggressive marketing by company 2 in differentiating the product offerings A and B, company 2 captures all the market, resulting in a payoff of $+1$ for company 2 and a payoff of -1 for company 1.

The two companies have to simultaneously decide (each one does not know the decision of the other) which product to produce. This is the strategic decision facing the two companies. This situation is captured by a strategic form game $\Gamma = \langle N, S_1, S_2, u_1, u_2 \rangle$, where $N = \{1, 2\}$; $S_1 = S_2 = \{A, B\}$, and the utility functions are as described in the above table.

4.3 Rock-Paper-Scissors Game

This is an example of another two player zero-sum game where each player has three strategies, called *rock*, *paper*, and *scissors*. Actually, this is a popular hand game played by two persons. Another name for this game is *roshambo*. Two players simultaneously display one of three symbols: a *rock*, a *paper*, or *scissors*. The *rock* symbol beats *scissors* symbol; *scissors* symbol beats *paper* symbol; *paper* symbol beats *rock* symbol (symbolically, rock can break scissors; scissors can cut paper; and paper can cover rock). The payoff matrix for this game is given as follows.

	2		
1	Rock	Paper	Scissors
Rock	$0, 0$	$-1, 1$	$1, -1$
Paper	$1, -1$	$0, 0$	$-1, 1$
Scissors	$-1, 1$	$1, -1$	$0, 0$

4.4 BOS (Bach or Stravinsky) Game

This game is named after the famous musicians Bach and Stravinsky. This is also called the *Battle of Sexes* game. This is a game where the players want to coordinate with each other; however, they have a disagreement about which of the outcomes is better. Let us say two players 1 and 2 wish to go out together to an event A or to an alternative event B. Player 1 prefers to go to event A and player 2 prefers to go to event B. The payoff matrix is as shown.

	2	
1	A	B
A	$2, 1$	$0, 0$
B	$0, 0$	$1, 2$

Clearly, this game captures a situation where the players want to coordinate but they have conflicting interests. The outcomes (A, B) and (B, A) are unfavorable to either player. The choice is essentially between (A, A) and (B, B). Recalling the company analogy, suppose we have two companies, 1 and 2. Each company can produce only one of two competing products A and B, but at any given time, a company can only produce one type of product. Assume product A is a niche product of company 1 while product B is a niche product of company 2. If both the companies produce product A, the consumers are compelled to buy product A and would naturally prefer to buy it from company 1 rather than from 2. Assume that company 1 captures two thirds of the market. We can reflect this fact by making the payoff to company 1 twice that of company 2. If both the companies produce product B, the reverse situation will prevail and company 2 will make twice as much payoff as company 1.

On the other hand, if the two companies decide to produce different products, then the market gets segmented and each company tries to outwit the other through increased spending on advertising. In fact, their competition may actually benefit a third company and, effectively, neither company 1 nor company 2 makes any payoff. The above table depicts the payoff structure for this game.

4.5 A Coordination Game

We have already presented this game in Chapter 1 (student's coordination game). This game is similar to the BOS game but the two players now have a preference for the same option, namely event A. The payoff matrix is as shown below; note that the outcomes (A, A) and (B, B) in that order are preferred.

	2	
1	A	B
A	$10, 10$	$0, 0$
B	$0, 0$	$1, 1$

Continuing our analogy of companies, the above game corresponds to a situation wherein the two companies produce the same product, and they have equal market share. This market share is ten times as much for product A as for product B. On the other hand, if the two companies produce different products, a third company may capture all the market share leaving nothing for companies 1 and 2.

Since the payoffs are the same for both the players in all outcomes, such games are also called *common payoff games*.

4.6 Prisoner's Dilemma Game

This is one of the most extensively studied problems in game theory, with many interesting interpretations in a wide variety of situations. Two individuals are arrested for allegedly committing a crime and are lodged in separate cells. The interrogator questions them separately. The interrogator privately tells each prisoner that if he is the only one to confess, he will get a light sentence of 1 year in jail while the other would be sentenced to 10 years in jail. If both players confess, they would get 5 years each in jail. If neither confesses, then each would get 2 years in jail. The interrogator also informs each prisoner what has been told to the other prisoner. Thus the payoff matrix is common knowledge.

	2	
1	NC	C
NC	$-2, -2$	$-10, -1$
C	$-1, -10$	$-5, -5$

How would the prisoners behave in such a situation? They would like to play a strategy that offers a best response to a best response strategy that the other player may adopt, the latter player also would like to play a strategy that offers a best response to the other player's best response strategy, and so on. First observe that

C is each player's best response strategy regardless of what the other player plays:

$$u_1(C,C) = -5 > u_1(NC,C) = -10; \quad u_1(C,NC) = -1 > u_1(NC,NC) = -2.$$
$$u_2(C,C) = -5 > u_2(C,NC) = -10; \quad u_2(NC,C) = -1 > u_2(NC,NC) = -2.$$

Thus (C,C) is a natural prediction for this game. However, the outcome (NC,NC) is the best outcome jointly for the players. Prisoner's dilemma is a classic example of a game where rational, intelligent behavior does not lead to an outcome where the sum of utilities of the players is maximal. Also, each prisoner has a negative effect or externality on the other. When a prisoner moves away from (NC,NC) to reduce his jail term by 1 year, the jail term of the other prisoner increases by 8 years.

An alternate way of interpreting the strategies of the prisoners is also popular. In this interpretation, each prisoner has two strategies, *cooperate* and *defect*. The strategy *cooperate* corresponds to cooperating with the other player by not confessing and therefore is equivalent to the strategy NC. The strategy *defect* corresponds to betraying the other player by confessing to the crime and therefore is equivalent to the strategy C. In the rest of the book, we will consistently use the C and NC strategies and not the *defect* and *cooperate* nomenclature.

4.7 Company's Dilemma Game

On the lines of the prisoner's dilemma problem, we present an analogous game involving two companies. Consider two companies 1 and 2, each of which can produce two competing products A and B, but only one at a time. The companies are known better for product A than for product B. However, environmentalists have launched a negative campaign on product A branding it as eco-unfriendly.

If both the companies produce product A, then, in spite of the negative campaign, their payoff is quite high since product A happens to be a niche product of both the companies. On the other hand, if both the companies produce product B, they still make some profit, but not as much as they would if they both produced product A.

On the other hand, if one company produces product A and the other company produces product B, then because of the negative campaign about product A, the company producing product A makes zero payoff while the other company captures all the market and makes a high payoff. The table below depicts the payoff structure for this game.

	2	
1	A	B
A	$6,6$	$0,8$
B	$8,0$	$3,3$

4.8 A Non-Symmetric Company's Dilemma Game

The examples we have provided so far, namely, matching pennies; rock-paper-scissors; BOS; coordination; prisoner's dilemma; and company's dilemma are instances of *symmetric games*. A two player strategic form game is called symmetric if $S_1 = S_2$ and $u_1(s_1, s_2) = u_2(s_2, s_1) \quad \forall s_1 \in S_1, \ \forall s_2 \in S_2$. We now provide an example of a non-symmetric game involving two competing companies 1 and 2. In this game also, each company has to simultaneously decide which of the two products A, B, it will produce. Company 1 is better known for product A and if it happens that both companies produce A, company 1 prospers. If both companies produce B, then they share the profits equally. If one of them produces A and the other produces B, then company 2 prospers (perhaps due to its more aggressive marketing). The following payoff matrix captures the strategic situation facing the two companies.

	2	
1	A	B
A	$4, 1$	$0, 4$
B	$1, 5$	$1, 1$

4.9 A Duopoly Pricing Game

This is due to Bertrand (1883). There are two companies 1 and 2 which wish to maximize their profits. The demand as a function of a price p is given by a continuous and strictly decreasing function $x(p)$. The cost for producing each unit of product is c where $c > 0$. The companies simultaneously choose their prices p_1 and p_2. The amount of sales for each company is given by:

$$\begin{aligned} x_1(p_1, p_2) &= x(p_1) && \text{if} \quad p_1 < p_2 \\ &= \tfrac{x(p_1)}{2} && \text{if} \quad p_1 = p_2 \\ &= 0 && \text{if} \quad p_1 > p_2 \end{aligned}$$

$$\begin{aligned} x_2(p_1, p_2) &= x(p_2) && \text{if} \quad p_2 < p_1 \\ &= \tfrac{x(p_2)}{2} && \text{if} \quad p_1 = p_2 \\ &= 0 && \text{if} \quad p_2 > p_1 \end{aligned}$$

It is assumed that the firms incur production costs only for an output level equal to their actual sales. Given prices p_1 and p_2, the utilities of the two companies are:

$$u_1(p_1, p_2) = (p_1 - c)\ x_1(p_1, p_2)$$
$$u_2(p_1, p_2) = (p_2 - c)\ x_2(p_1, p_2)$$

Note that for this game, $N = \{1, 2\}$ and $S_1 = S_2 = [0, \infty)$. This is an infinite game since the strategy sets are infinite.

4.10 Tragedy of the Commons

The *Tragedy of the Commons* represents a type of social paradox or social tragedy. The problem involves a conflict over use of resources between individual interests and social interests. A village has n farmers represented by the set $N = \{1, 2, \ldots, n\}$. Each farmer has the option of keeping a sheep or not. If 1 corresponds to keeping a sheep and 0 corresponds to not keeping a sheep, the strategy sets are given by

$$S_1 = S_2 = \cdots = S_n = \{0, 1\}.$$

The utility from keeping a sheep (that arises because of milk, wool, etc.) is equal to 1 unit. The village has a limited stretch of grassland and when a sheep grazes on this, there is a damage to the environment, equal to 5 units. This damage to the environment is to be shared equally by the farmers.

Let s_i be the strategy of each farmer. Then $s_i \in \{0, 1\}$. The payoff to farmer i is given by:

$$u_i(s_1, \ldots, s_i, \ldots, s_n) = s_i - \left[\frac{5(s_1 + \cdots + s_n)}{n}\right]$$

For the case $n = 2$, the payoff matrix would be:

	2	
1	0	1
0	$0, 0$	$-2.5, -1.5$
1	$-1.5, -2.5$	$-4, -4$

If $n > 5$, a farmer gains more utility by keeping a sheep rather than not having one. If $n < 5$, then the farmer gets less utility by keeping a sheep than not having one. If $n = 5$, the farmer can be indifferent between keeping a sheep and not keeping a sheep.

If the Government now imposes a pollution tax of 5 units to every farmer keeping a sheep, the payoff becomes:

$$u_i(s_1, \ldots, s_i, \ldots, s_n) = s_i - 5s_i - \frac{5(s_1 + \cdots + s_n)}{n}$$

Now every farmer will prefer not to keep a sheep. We will be analyzing this game in Chapters 5 and 6 to gain insights into this social situation.

4.11 Bandwidth Sharing Game

This problem is based on an example presented by Tardos and Vazirani [2]. There is a shared communication channel of maximum capacity 1. There are n users of this channel, and user i wishes to send x_i units of flow, where $x_i \in [0, 1]$. We have

$$N = \{1, 2, \ldots, n\}$$
$$S_1 = S_2 = \ldots = S_n = [0, 1].$$

If $\sum_{i \in N} x_i \geq 1$, then the transmission cannot happen since the capacity is exceeded, and the payoff to each player is zero. If $\sum_{i \in N} x_i < 1$, then assume that the following is the payoff to user i:

$$u_i(x_1, \ldots, x_n) = x_i\big(1 - \sum_{j \in N} x_j\big) \;\; \forall i \in N.$$

The above expression models the fact that the payoff to a player is proportional to the flow sent by the player but is negatively impacted by the total flow. The second term captures the fact that the quality of transmission deteriorates with the total bandwidth used. Note that the above is an infinite game (since the strategy sets are real intervals). We will show in Chapter 6 that an equilibrium outcome here is not socially optimal.

4.12 A Sealed Bid Auction

There is a seller who wishes to allocate an indivisible item to one of n prospective buyers in exchange for a payment. Here, $N = \{1, 2, \ldots, n\}$ represents the set of buying agents. Let $v_1, v_2, \ldots, v_n$ be the valuations of the players for the object. The n buying agents submit sealed bids and these bids need not be equal to the valuations. Assume that the sealed bid from player i ($i = 1, \ldots, n$) could be any real number greater than 0. Then the strategy sets of the players are: $S_i = (0, \infty)$ for $i = 1, \ldots, n$. Assume that the object is awarded to the agent with the lowest index among those who bid the highest. Let $b_1, \ldots, b_n$ be the bids from the n players. Then the allocation function will be:

$$y_i(b_1, \ldots, b_n) = \begin{cases} 1 & \text{if } b_i > b_j \text{ for } j = 1, \ldots, i-1; \quad b_i \geq b_j \text{ for } j = i+1, \ldots, n \\ 0 & \text{otherwise.} \end{cases}$$

In the first price sealed bid auction, the winner pays an amount equal to his bid, and the losers do not pay anything. In the second price sealed bid auction, the winner pays an amount equal to the highest bid among the players who do not win, and as usual the losers do not pay anything. The payoffs or utilities to the bidders in these two auctions are of the form:

$$u_i(b_1, \ldots, b_n) = y_i(b_1, \ldots, b_n)(v_i - t_i(b_1, \ldots, b_n))$$

where $t_i(b_1, \ldots, b_n)$ is the amount to be paid by bidder i in the auction when player i bids b_i $(i = 1, \ldots, n)$. Assume that $n = 4$, and suppose the valuations are $v_1 = 20$; $v_2 = 20$; $v_3 = 16$; $v_4 = 16$, and the bids are $b_1 = 10$; $b_2 = 12$; $b_3 = 8$; $b_4 = 14$. Then for both first price and second price auctions, we have the allocation $y_1(b) = 0$; $y_2(b) = 0$; $y_3(b) = 0$; $y_4(b) = 1$, where $b = (b_1, b_2, b_3, b_4)$. The payments for the first price auction are $t_1(b) = 0$; $t_2(b) = 0$; $t_3(b) = 0$; $t_4(b) = 14$ whereas the payments for the second price auction would be: $t_1(b) = 0$; $t_2(b) = 0$; $t_3(b) = 0$; $t_4(b) = 12$. The utilities can be easily computed from the valuations and the payments.

4.13 Pigou's Network Game

This example captures the effect of selfish routing when strategic players act independently of one another. A directed graph consists of two nodes S and T and there are two disjoint edges A and B connecting S to T (see Figure 4.1). A certain amount of traffic has to move from S to T. Each edge is associated with a cost function $c(\cdot)$ which describes the cost (for example travel time) from S to T, incurred by the users of that edge, as a function of the fraction of total traffic that is routed on that edge. Suppose $x \in [0, 1]$ denotes the fraction of traffic routed. On the edge A, the cost function is $c(x) = x \ \forall x \in [0, 1]$. On the edge B, the cost function is constant and equal to unity, that is, $c(x) = 1 \ \forall x \in [0, 1]$. A routing network of the above type is called a Pigou network [3].

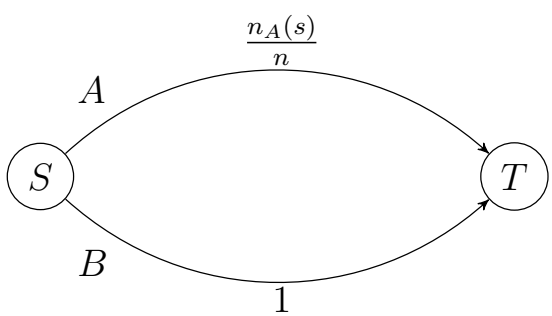

Fig. 4.1: A Pigou network

We shall consider a simple, stylized, discrete version of the above network with just two users, that is, $N = \{1, 2\}$. Each user has two strategies A and B corresponding to the two routes she may select. Thus we have $S_1 = S_2 = \{A, B\}$. Each user selects a route to be followed simultaneously and independent of the other user. A natural way of defining the payoff of a user for a strategy profile here would be negative of the cost of travel of the user. For example, suppose the strategy profile is (A, B) (player 1 selecting route A and player 2 selecting route B). The cost of

travel for player 1 would be $\frac{1}{2}$ since the fraction of traffic routed through edge A is $\frac{1}{2}$. The payoff for player 1 becomes $-\frac{1}{2}$. The cost of travel for player 2 is 1 because the edge B is selected and the payoff becomes -1. The payoff matrix for this simple routing game is therefore given by

	2	
1	A	B
A	$-1, -1$	$-\frac{1}{2}, -1$
B	$-1, -\frac{1}{2}$	$-1, -1$

If we have n users, we have $N = \{1, \ldots, n\}$ and $S_1 = \ldots = S_n = \{A, B\}$. Let $s = (s_1, \ldots, s_n)$ be a strategy profile chosen by all the users. To define the payoff function, we shall first define $n_A(s)$ as the number of players who have chosen the route A in the strategy profile s. Similarly, $n_B(s)$ denotes the number of users who have chosen route B in the strategy profile s. Clearly, $n_A(s) + n_B(s) = n$ for all strategy profiles s. The payoff function is given by

$$\begin{aligned} u_i(s) &= -\frac{n_A(s)}{n} \quad && \text{if } \ s_i = A \\ &= -1 && \text{if } \ s_i = B \end{aligned}$$

We will analyze the above game in Chapter 6.

4.14 Braess Paradox Game

This game is developed on the lines of the game presented in the book by Easley and Kleinberg [4]. This game illustrates the Braess paradox which is named after the German mathematician Dietrich Braess. This paradox is usually associated with transportation networks and brings out the counter-intuitive fact that a transportation network with extra capacity added may actually perform worse (in terms of time delays) than when the extra capacity did not exist. Figure 4.2 shows a network that consists of a source S and a destination T, and two intermediate hubs A and B. It is required to travel from S to T. One route is via the hub A and the other route proceeds via the hub B.

Regardless of the number of vehicles on the route, it takes 25 minutes to travel from S to B or from A to T. On the other hand, the travel time from S to A takes time $\frac{m}{50}$ minutes where m is the number of vehicles on that link. Similarly, the travel time from B to T takes time $\frac{m}{50}$ minutes where m is the number of vehicles on that link. Assume that there are $n = 1000$ vehicles that wish to move from S to T. This means $N = \{1, 2, \ldots, 1000\}$. The strategy sets are $S_1 = \ldots = S_n = \{A, B\}$. Given a strategy profile $(s_1, \ldots, s_n)$, let $n_A(s_1, \ldots, s_n)$ $(n_B(s_1, \ldots, s_n))$ denote the number of vehicles taking the route via A (B). It is easy to note that

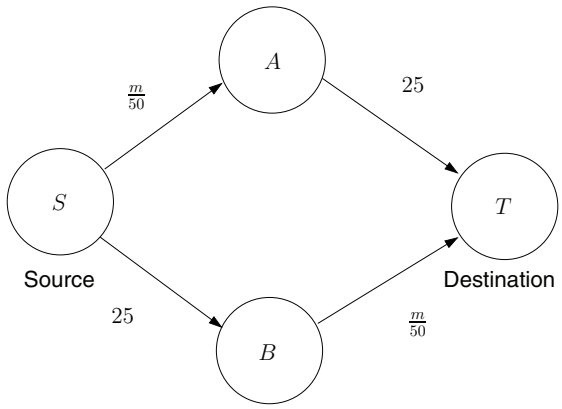

Fig. 4.2: A transportation network with four nodes

$n_A(s_1, \ldots, s_n) + n_B(s_1, \ldots, s_n) = n$. Since we wish to minimize travel time from S to T, it is convenient to define the utility of a player (in this case vehicle) as the negative of the travel time for that player from S to T. It is easy to see that

$$\begin{aligned} u_i(s_1, \ldots, s_n) &= -25 - \frac{n_A(s_1, \ldots, s_n)}{50} \quad \text{if } s_i = A \\ &= -25 - \frac{n_B(s_1, \ldots, s_n)}{50} \quad \text{if } s_i = B \end{aligned}$$

This defines a strategic form game. Note that

$$u_i(A, A, \ldots, A) = u_i(B, B, \ldots, B) = -45$$

$$u_i(s_1, s_2, \ldots, s_n) = -35 \quad \text{whenever } n_A(s_1, \ldots, s_n) = n_B(s_1, \ldots, s_n) = 500$$

Let us say we now introduce a fast link from A to B to ease the congestion in the network (as a degenerate case, we will assume the travel time from A to B to be zero). Figure 4.3 depicts this new network with extra capacity added from A to B. Now the strategies available to each vehicle are to go from S to A to T (call this strategy A); S to B to T (call this strategy B); and S to A to B to T (call this strategy AB). So we have $S_1 = \ldots = S_n = \{A, B, AB\}$. Defining $n_A(s_1, \ldots, s_n), n_B(s_1, \ldots, s_n), n_{AB}(s_1, \ldots, s_n)$ on the same lines as before, we get

$$\begin{aligned} u_i(s_1, \ldots, s_n) &= -25 - \frac{n_A(s_1, \ldots, s_n) + n_{AB}(s_1, \ldots, s_n)}{50} \quad \text{if } s_i = A \\ &= -25 - \frac{n_B(s_1, \ldots, s_n) + n_{AB}(s_1, \ldots, s_n)}{50} \quad \text{if } s_i = B \\ &= -\frac{n_A(s_1, \ldots, s_n) + n_{AB}(s_1, \ldots, s_n)}{50} \\ &\quad -\frac{n_B(s_1, \ldots, s_n) + n_{AB}(s_1, \ldots, s_n)}{50} \quad \text{if } s_i = AB \end{aligned}$$

We will analyze the above two games in Chapters 5 and 6 and illustrate the Braess paradox.

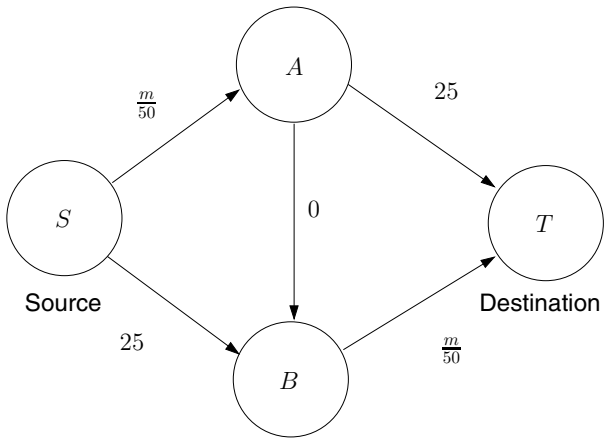

Fig. 4.3: Transportation network with an extra link from A to B

4.15 Summary and References

In this chapter, we presented many examples of strategic form games (also called normal form games) to gain an appreciation of how real world strategic situations could be abstracted as games. The examples we presented include:

- *Matching Pennies*, a two person zero-sum game and *Rock-Paper-Scissors* game, another two person zero-sum game. The games are called zero-sum because the sum of the utilities of the two players is zero in every outcome of the game. These are also called *strictly competitive games*.
- *BOS (Bach or Stravinsky) game*, also called the battle of the sexes game, captures a two player situation where the two players wish to coordinate with each other but they have different preferences for outcomes.
- *Coordination game* or the student's coordination game where two players derive a positive payoff only when they are together, however the payoff they derive when they are together depends on which actions they select.
- *Prisoner's Dilemma*, a classic two player game which illustrates many nuances of strategic conflict and cooperation. This is a game where rational and intelligent behavior does not lead to a socially optimal outcome.
- *Company's Dilemma* which shows how the prisoner's dilemma can be used to model a strategic situation facing two competing companies trying to outwit each other.
- *Non-Symmetric Company's Dilemma* which captures the strategic situation facing two companies which do not have a symmetric payoff structure.
- *Duopoly Pricing Game* which models the strategic situation facing two competing companies in deciding the price at which to sell a particular product.

- *Tragedy of the Commons*, a well studied problem that illustrates a common social paradox involving a conflict over resource sharing. This is a multi-person game.
- *Bandwidth Sharing* game, a multi-person game with infinite strategy sets which illustrates the conflict that arises due to sharing of a common resource.
- *Sealed Bid Auction*, which introduces the strategic form games underlying the well known first price auction and second price auction.
- *Pigou's Network* game which illustrates the notion of selfish routing by strategic agents acting independently of one another
- *Braess Paradox* game which faithfully captures the strategic conflict leading to the famous paradox in routing games.

The above examples are fairly representative of typical situations discussed in this book. We state cautiously that there are numerous other interesting and popular examples that we have left out in our discussion – in fact, we have covered only a minuscule subset of popular examples here. The following books also contain many illustrative examples of strategic form games: Osborne [1], Straffin [5], and Binmore [6], and Maschler, Solan, and Zamir [7].

The material discussed in this chapter draws upon mainly from three sources, namely the books by Myerson [8], Mascolell, Whinston, and Green [9], and Osborne and Rubinstein [1]. The paper by Tardos and Vazirani [2] is a fine introduction to concepts in game theory; we have taken many examples from their paper.

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[4] David Easley and Jon Kleinberg. *Networks, Crowds, and Markets: Reasoning About a Highly Connected World.* Cambridge University Press, 2010.

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4.16 Exercises

(1) There are many interesting games that we have not discussed in this chapter. Explore other examples such as hawk and dove (also called chicken), cold war, pollution control, Cournot pricing game, ISP routing game, available from the literature, for example, the following books [10, 11, 5, 6, 7].

(2) There are n players. Each player announces a number in the set $\{1, ..., K\}$, where K is a fixed positive integer. A prize of \$1 is split equally between all the people whose number is closest to $\frac{2}{3}$ of the average number. Formulate this as a strategic form game.

(3) Develop the strategic form game for the Pigou network game for $n = 3$ and $n = 4$.

(4) Consider the following strategic form game (network formation game). The nodes of the network are the players: $N = \{1, 2, \ldots, n\}$. The strategy set S_i of player i is the set of all subsets of $N \setminus \{i\}$. A strategy of a node is to decide on with which other nodes it would like to have links. A strategy profile corresponds to a particular network or graph. Assume that δ where $0 < \delta < 1$ is the benefit that accrues to each node of a link while $c > 0$ is the cost to each node of maintaining the link. Further, assume that δ^k is the benefit that accrues from a k-hop relationship, where, k is the length of a shortest path between the two involved nodes. A link is formed under mutual consent while it can be broken unilaterally. Given a graph g formed out of a strategy profile, let the utility $u_i(g)$ of node i be given by

$$u_i(g) = \sum_{j \neq i} \delta^{l_{ij}(g)} - c.d_i(g)$$

where $l_{ij}(g)$ is the number of links in a shortest path between i and j and $d_i(g)$ is the degree of node i. Write down the strategy profiles corresponding to following structures of graphs assuming n nodes and compute the utilities of all individual nodes.

- Complete graph
- Straight line graph
- Star graph